BASED ON LATEST OCTOBER PATTERN

Infosys Pseudo Code Questions

Topic: Do-While Loop Execution & Logic

runs atleast once because condition is at the end.

Question 01

```
Set Integer m=10
do m=m*2
display m++
while(m<1)
end do-while
```

A) 11 22

B) 20

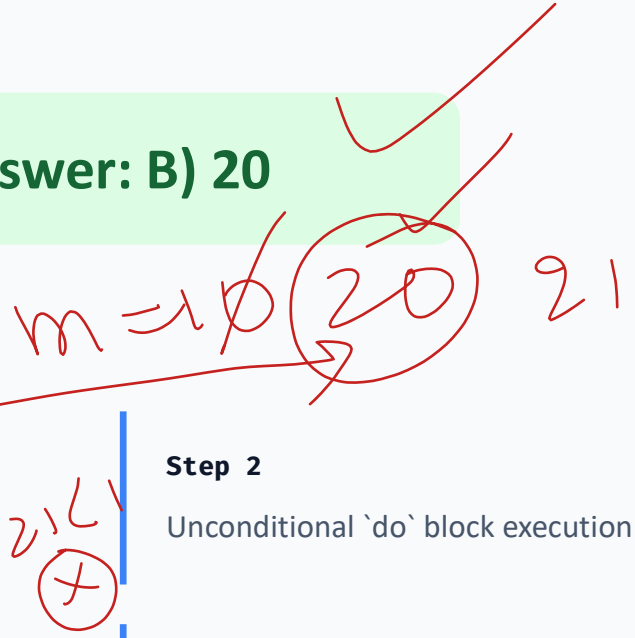
C) Infinite loop

D) 10 20

Answer: B) 20

Execution Dry Run

```
Set Integer m=10
do m=m*2
display m++
while(m<1)
end do-while
```



Step 1

Initialization: **m = 10**.

Step 2

Unconditional `do` block execution: **m = 10 * 2 = 20**.

Step 3

`display m++` prints **20**, then increments `m` to **21**.

Step 4

Condition `while(21 < 1)` is **False**. Loop terminates.

Question 02

```
Set Integer x = 2, sum = 0
do
if (x MOD 3 == 0)
then do
sum = sum + x
end-if
x = x + 2
while(x < 15)
display sum
```

A) 18

B) 24

C) 30

D) 12

```
Set Integer x = 2, sum = 0
do
if (x MOD 3 == 0)
then do
sum = sum + x
end-if
x = x + 2
while(x < 15)
display sum
```

Answer: A) 18

x=2: `2 MOD 3 != 0`, sum=0, x=4.

x=4: `4 MOD 3 != 0`, sum=0, x=6.

x=6: `6 MOD 3 == 0`, sum=6, x=8.

x=8: `8 MOD 3 != 0`, sum=6, x=10.

x=10: `10 MOD 3 != 0`, sum=6, x=12.

x=12: `12 MOD 3 == 0`, sum=18, x=14.

x=14: `14 MOD 3 != 0`, sum=18, x=16.

End: `16 < 15` is **False**. Output: 18.

Question 03

```
Set Integer a = 5, b = 3
do
a = a ^ b
b = b + 1
while(b < 6)
display a
```

A) 5

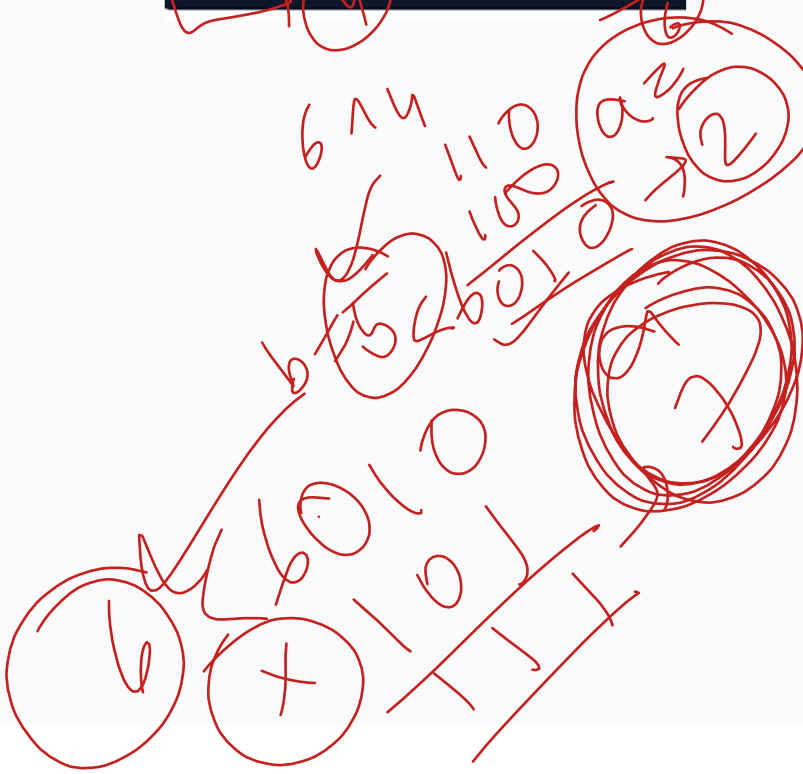
B) 2

C) 7

D) 4

Answer: C) 7

```
Set Integer a = 5, b = 3
do
a = a ^ b
b = b + 1
while(b < 6)
display a
```



Iteration 1 (b=3)

`a = 5 ^ 3` -> Binary: `101` XOR `011` = `110` (Dec 6). `b` becomes 4. `4 < 6` is **True**.

Iteration 2 (b=4)

`a = 6 ^ 4` -> Binary: `110` XOR `100` = `010` (Dec 2). `b` becomes 5. `5 < 6` is **True**.

Iteration 3 (b=5)

`a = 2 ^ 5` -> Binary: `010` XOR `101` = `111` (Dec 7). `b` becomes 6. `6 < 6` is **False**.

Question 04

```
Set Integer p = 10, count = 0
do
p = p - 2
if (p == 6)
then do
continue
end-if
count = count + 1
while(p > 0)
display count
```

A) 3

B) 4

C) 5

D) 6

```
Set Integer p = 10, count = 0
do
p = p - 2
if (p == 6)
then do
continue
end-if
count = count + 1
while(p > 0)
display count
```

Answer: B) 4

p=8: `count` becomes 1. `8 > 0` is True.

p=6: `p==6` is True. `continue` skips increment. `6 > 0` is True.

p=4: `count` becomes 2. `4 > 0` is True.

p=2: `count` becomes 3. `2 > 0` is True.

p=0: `count` becomes 4. `0 > 0` is False.

Loop Ends: Final count is 4.

Question 05

```
Set Integer k = 7, res = 1
do
res = res * k
k = k - 1
while(k > 10)
display res + " " + k
```

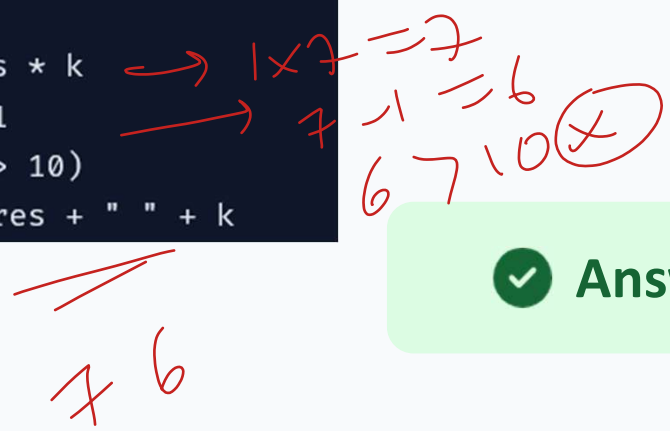
A) 7 6

B) 1 7

C) 42 5

D) 7 7

```
Set Integer k = 7, res = 1
do
res = res * k
k = k - 1
while(k > 10)
display res + " " + k
```



Answer: A) 7 6

1. Unconditional first execution.
2. `res = 1 * 7 = 7`.
3. `k = 7 - 1 = 6`.
4. `while(6 > 10)` is False.
5. **Outputs 7 and 6.**

Question 06

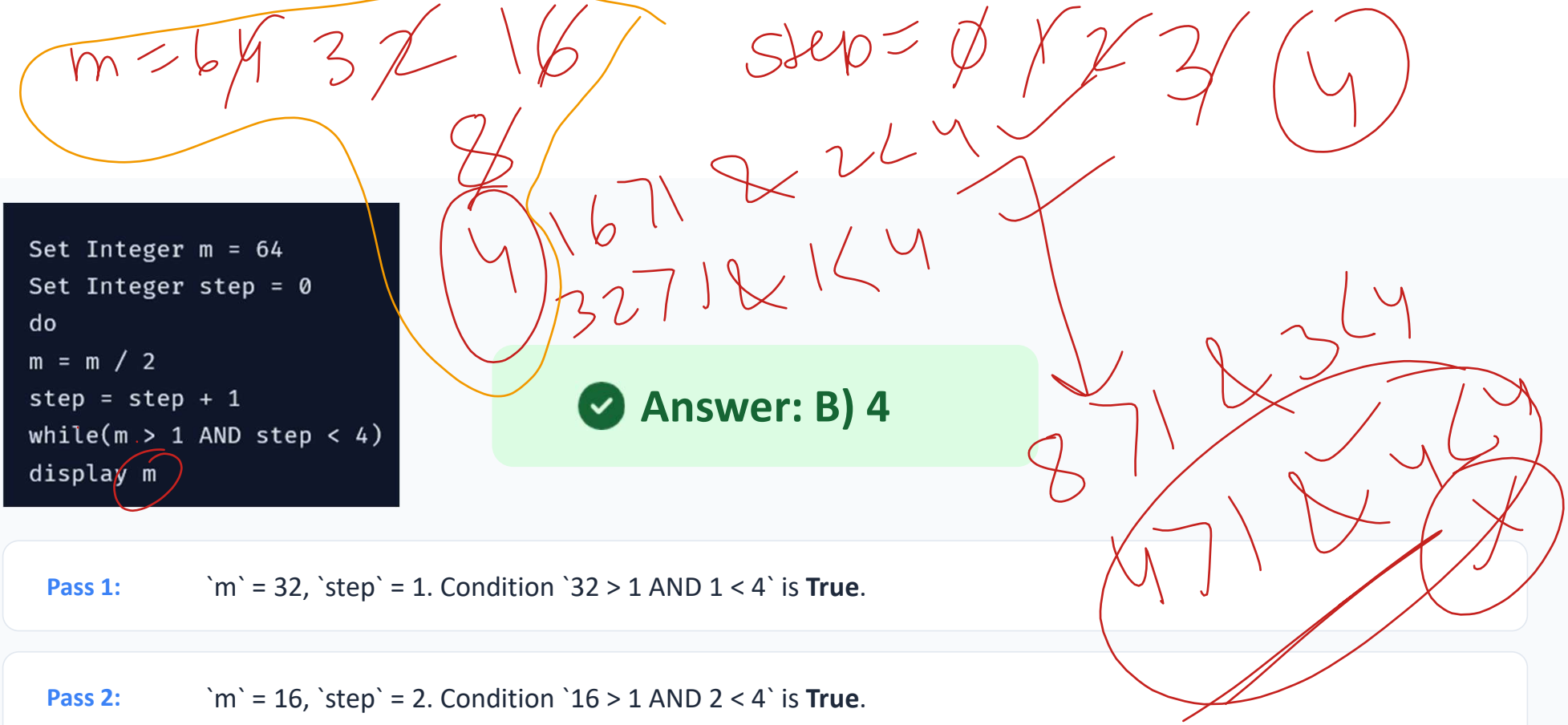
```
Set Integer m = 64
Set Integer step = 0
do
m = m / 2
step = step + 1
while(m > 1 AND step < 4)
display m
```

A) 8

B) 4

C) 2

D) 1



```
Set Integer m = 64
Set Integer step = 0
do
m = m / 2
step = step + 1
while(m > 1 AND step < 4)
display m
```

Answer: B) 4

Pass 1: `m` = 32, `step` = 1. Condition `32 > 1 AND 1 < 4` is **True**.

Pass 2: `m` = 16, `step` = 2. Condition `16 > 1 AND 2 < 4` is **True**.

Pass 3: `m` = 8, `step` = 3. Condition `8 > 1 AND 3 < 4` is **True**.

Pass 4: `m` = 4, `step` = 4. Condition `4 > 1 AND 4 < 4` is **False**.

Question 07

```
Set Integer arr[4] = {1, 3, 5, 7}
Set Integer i = 0, sum = 0
do
sum = sum + arr[i]
i = i + 2
while(i < 4)
display sum
```

A) 4

B) 6

C) 9

D) 16

Answer: B) 6

```
Set Integer arr[4] = {1, 3, 5, 7}
Set Integer i = 0, sum = 0
do
sum = sum + arr[i]
i = i + 2
while(i < 4)
display sum
```

Iteration 1

`i=0`. `sum = 0 + arr[0]` -> 0 + 1 = 1. `i` becomes 2. Condition `2 < 4` is **True**.

Iteration 2

`i=2`. `sum = 1 + arr[2]` -> 1 + 5 = 6. `i` becomes 4. Condition `4 < 4` is **False**.

Loop Ends

Total `sum` displayed is **6**.

Question 08

```
Set Integer a = 1, b = 20
do
a = a * 2
b = b - 5
while(a < b)
display a + b
```

A) 10

B) 11

C) 13

D) 14

```
Set Integer a = 1, b = 20
do
a = a * 2
b = b - 5
while(a < b)
display a + b
```



Answer: C) 13

Iteration 1: `a` = 1*2 = 2. `b` = 20-5 = 15.
Condition `2 < 15` is True.

Iteration 2: `a` = 2*2 = 4. `b` = 15-5 = 10.
Condition `4 < 10` is True.

Iteration 3: `a` = 4*2 = 8. `b` = 10-5 = 5.
Condition `8 < 5` is False.

Calculates `a + b` -> 8 + 5 = 13.

Question 09

```
Set Integer n = 5
Set Integer ans = 1
do
ans = ans + (n * 2)
n = n - 2
while(n >= 0)
display ans
```

A) 17

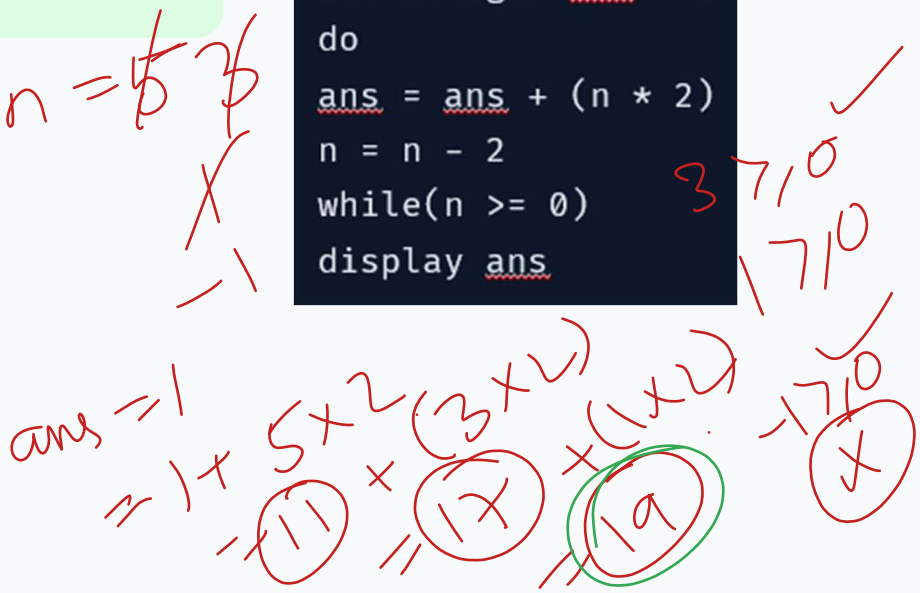
B) 19

C) 21

D) 11

Answer: B) 19

```
Set Integer n = 5
Set Integer ans = 1
do
ans = ans + (n * 2)
n = n - 2
while(n >= 0)
display ans
```



Iteration 1

`ans` = 1 + (5 * 2) = 11. `n` = 5 - 2 = 3. Condition `3 >= 0` is **True**.

Iteration 2

`ans` = 11 + (3 * 2) = 17. `n` = 3 - 2 = 1. Condition `1 >= 0` is **True**.

Iteration 3

`ans` = 17 + (1 * 2) = 19. `n` = 1 - 2 = -1. Condition `-1 >= 0` is **False**.

Question 10

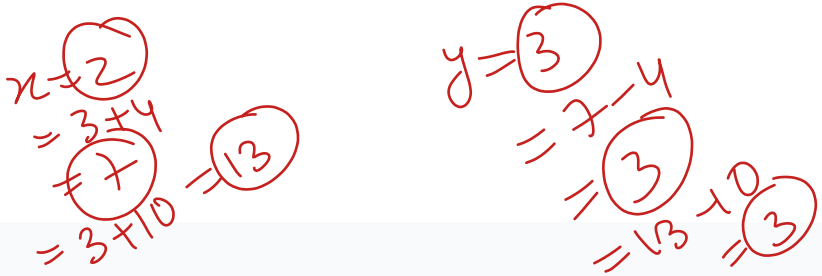
```
Set Integer x = 2, y = 3, z = 4
do
x = y + z
y = x - z
z = x + y
while(z < 15)
display x + y + z
```

A) 30

B) 32

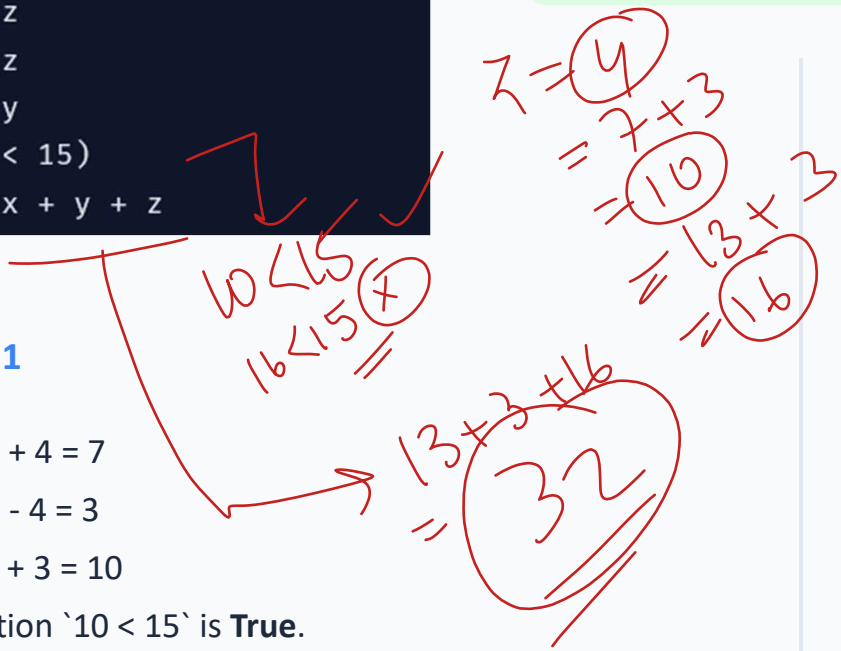
C) 24

D) 28



```
Set Integer x = 2, y = 3, z = 4
do
x = y + z
y = x - z
z = x + y
while(z < 15)
display x + y + z
```

Answer: B) 32



Iteration 1

- `x` = 3 + 4 = 7
- `y` = 7 - 4 = 3
- `z` = 7 + 3 = 10
- Condition `10 < 15` is **True**.

Iteration 2

- `x` = 3 + 10 = 13
- `y` = 13 - 10 = 3
- `z` = 13 + 3 = 16
- Condition `16 < 15` is **False**.

Output: 13 + 3 + 16 = 32.



End of Practice Set

Good luck with your Infosys technical rounds!